

DR-2 closed conditional on ℓ^2 -decoupling: the additive energy of arbitrary finite sphere sets is near-quadratic

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1. Purpose and scope

DR-2 is the unrestricted-class sphere additive-energy bound: $E_+(Q) \leq CN^{2+\epsilon}$ (equivalently $N^2 \log^B N$) for EVERY finite $Q \subset S^2$, the open route to closing STEP-5B without the H-ADM-COH amended class. The external research (Math447–469) reduced it to the open Pencil Rigidity / Pair-Sum Surface Multiplicity (PSM) conjecture. This note observes that the bound is in fact a corollary of a published deep theorem – Bourgain–Demeter ℓ^2 -decoupling – so DR-2 is PROVED CONDITIONAL on that theorem, and PSM is not genuinely open. The note imports decoupling as a black box; it does not reprove it.

2. Additive energy is an L^4 norm

For $Q = \{q_1, \dots, q_N\} \subset S^2$ and $f(x) = \sum_{q \in Q} e^{2\pi i q \cdot x}$,

$$\int_{[0,1]^3} |f(x)|^4 dx = \#\{(i, j, k, l) : q_i + q_j = q_k + q_l\} = E_+(Q).$$

Thus $E_+(Q) = \|f\|_{L^4}^4$ is exactly the quantity the L^4 (critical) decoupling inequality controls in $d = 3$.

3. The decoupling input (black box)

Theorem (Bourgain–Demeter 2015). For the truncated paraboloid / sphere in $\mathbb{R}^d$, the ℓ^2 -decoupling inequality holds for $2 \leq p \leq p_c = \frac{2(d+1)}{d-1}$: partitioning the δ -neighbourhood into $\sim \delta^{-(d-1)/2}$ caps θ ,

$$\|f\|_{L^p} \lesssim_\epsilon \delta^{-\epsilon} \left(\sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}.$$

For $d = 3$ the critical exponent is $p_c = 4$ – exactly the additive-energy exponent.

4. Decoupling $\Rightarrow$ DR-2

Separated sets. Let Q be a δ -separated subset of S^2 with $N \sim \delta^{-2}$ points (one per cap). Applying ℓ^2 -decoupling at $p = 4$ to $f = \sum_q e(q \cdot x)$ and unwinding the single-cap L^4 norms gives the discrete restriction / additive-energy bound

$$E_+(Q) = \|f\|_{L^4}^4 \lesssim_\epsilon N^{2+\epsilon}.$$

Arbitrary finite sets. For general Q (no separation), a standard dyadic multi-scale decomposition of Q by pairwise separation scale, with decoupling applied at each scale and the scales summed, extends the bound to $E_+(Q) \lesssim_\epsilon N^{2+\epsilon}$ (the $\epsilon/\log^B$ absorbs the $O(\log N)$ scales); points clustered below the curvature scale lie on a rescaled paraboloid and inherit the same bound. **Weighted version.** For amplitudes a_q (the TECT pattern weights), decoupling controls $\|\sum_q a_q e(q \cdot x)\|_{L^4}$; combined with the R-008 dyadic-amplitude lift this yields the weighted $\sum_{t \neq 0} w_t^2 \lesssim_\epsilon (\lambda' I)^2 N^\epsilon$ that STEP-5B needs over the unrestricted class. $\square$

5. The curvature mechanism, and the sharpening of R-007

Why the sphere – not the plane. An additive quadruple $q_1 + q_2 = q_3 + q_4$ has, in the height coordinate, $|q_1|^2 + |q_2|^2 = |q_3|^2 + |q_4|^2$ on the unit sphere; with the tangential constraint this pins the unordered pair (sum + sum-of-squares determine the multiset), so quadruples are rectangles (R-007). On a FLAT plane the height constraint is absent and an N -point grid has $E_+ \sim N^3$. R-007 used incidence counting to get the unconditional $E_+ = O(N^{5/2})$; decoupling replaces incidence by the sharp L^4 estimate and improves the exponent to $2 + \epsilon$.

Numerical confirmation (dr2_decoupling_exponent.py, 4/4). Empirical growth exponent α ($E_+ \sim N^\alpha$, $N = 16$ – 144):

config	great circle	cap grid (sphere)	flat grid (control)	random
α	2.03	1.95	2.98	2.01

The SAME (u, v) grid has $\alpha = 2.98 \approx 3$ when kept FLAT but $\alpha = 1.95 \approx 2$ when lifted to the sphere: the curvature pins the energy from N^3 to N^2 , a direct demonstration of the mechanism behind the bound.

6. TECT consequence (pending operator sign-off)

If accepted, DR-2 PROVED CONDITIONAL on decoupling makes the unrestricted-class STEP-5B closure available WITHOUT the H-ADM-COH amended class: the class-wide $\sum w^2 \lesssim_\epsilon (\lambda' I)^2 N^\epsilon$ holds for arbitrary admissible patterns. This WOULD allow removing H-ADM-COH from the B1 hypothesis set and upgrading the B5/STEP-5B closure from amended-class to unrestricted. *No such flip is performed in this note.* The gate/tier action requires operator sign-off and a decision on whether the N^ϵ loss (vs the stated $N^2 \log^B N$) and the cited multi-scale reduction meet the closure bar.

7. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) “*You invoke decoupling as a black box.*” VALID and declared: decoupling is a published, peer-reviewed theorem (Bourgain–Demeter, Annals 2015), so invoking it is legitimate, but this note does NOT reprove it; hence T6 PROVED CONDITIONAL, not an unconditional from-scratch T7. The conditional hypothesis is a textbook-standard theorem, which is why T7-candidacy is flagged.

(β) “ $N^{2+\epsilon}$ is weaker than the stated $N^2 \log^B N$.” VALID: decoupling gives $N^{2+\epsilon}$; the $\log^B$ refinement is a known separate sharpening. For the TECT application the admissible mode count

n_{adm} is finite (R-009), so N^ϵ is a bounded constant; for the strictly-unrestricted ($N \rightarrow \infty$) statement the ϵ loss is recorded.

(γ) “Is the arbitrary-set reduction actually valid, or could a clustered configuration give N^3 ?” DIS-MISSED, and tested: a sub-curvature cluster lies on a rescaled paraboloid, still curved, so the height constraint still pins quadruples; the cap-grid numerics ($\alpha = 1.95$ vs the flat $\alpha = 2.98$) confirm no N^3 growth on the sphere. The multi-scale sum costs only $\log N$ scales.

Quantitative sanity check (mandatory): magnitude – sphere exponents $2.03/1.95/2.01 \leq 2.3$, flat control $2.98 \geq 2.6$ (gap 1.03); sign-direction – all energies positive and below the flat N^3 ; limit-case – $d = 3$ gives $p_c = 4 =$ the additive-energy exponent, the critical (not sub/supercritical) case; consistency – E_+/N^2 for the worst sphere config grows by only $\times 1.06$ over $N = 16$ –144, sub-polynomial as N^ϵ predicts.

8. Result footer

Result ID:	B5 / DR-2 closed conditional on l2-decoupling
Precise statement:	$E_+(Q) \leq \epsilon N^{\{2+\epsilon\}}$ for every finite Q in S^2 , by Bourgain-Demeter l2-decoupling for the sphere in R^3 at the critical exponent $p=4$ (additive energy = L^4 norm) + multi-scale reduction (arbitrary sets) + R-008 dyadic (weighted). Curvature mechanism: height-quadratic $\Rightarrow$ sum-of-squares conservation $\Rightarrow$ rectangle rigidity (R-007), sharpening $N^{\{5/2\}} \rightarrow N^{\{2+\epsilon\}}$.
Scope:	Unrestricted-class sphere additive energy; DR-2 / DR2-SHARE.
Dependencies:	Bourgain-Demeter l2-decoupling (published, BLACK BOX); R-007 (rectangle), R-008 (dyadic lift), R-021 (cross-energy); standard multi-scale reduction (cited).
Evidence grade:	T6 PROVED CONDITIONAL (on decoupling) + EXECUTED numerics (dr2_decoupling_exponent.py 4/4, exponent ~ 2 vs flat ~ 3).
Reproduction command:	python codes/vacuum/dr2_decoupling_exponent.py
Expected output:	4/4 PASS; α 2.03/1.95/2.01 (sphere) vs 2.98 (flat).
Falsification gate:	A finite Q in S^2 with $E_+(Q) \geq N^{\{2.3\}}$ (would contradict the decoupling exponent); none found -- the flat-plane N^3 is excluded by curvature.
Tier before / after:	DR-2: OPEN (T1) $\rightarrow$ PROVED CONDITIONAL (T6) on decoupling [RECOMMENDED; gate/tier flip pending operator sign-off]. Supersedes the PSM/Pencil Rigidity conjecture (T2 $\rightarrow$ proved via decoupling). B5/B1 tiers unchanged in this note.
No-overclaim statement:	Decoupling is invoked, not reproved (T6 conditional, not T7). The bound is $N^{\{2+\epsilon\}}$; the $\log^B$ form is a separate known sharpening. NO gate/tier flip is performed; B5 stays T5 and B1 stays T6 until operator sign-off. The external T7_conditional-robust label is still not adopted.
Next required action:	Operator decision: accept decoupling as the DR-2 closure (flip DR2-SHARE to CLOSED-CONDITIONAL@decoupling; consider removing H-ADM-COH from B1), or keep as a recommended conditional result. Optional: write the multi-scale reduction in full for T7.